

Discrete Structures

Semester III — Detailed Unit-wise Notes (Course Code: 2418303)

Unit 1.0 — Logic and Proof Techniques

1.1 Propositional Logic

A **proposition** is a declarative statement that is either true (T) or false (F), but not both.

bullet **Connectives:**

bullet **Negation ($\neg p$)** — "not p", true when p is false.

bullet **Conjunction ($p \wedge q$)** — "p and q", true only when both p and q are true.

bullet **Disjunction ($p \vee q$)** — "p or q", true when at least one of p, q is true.

bullet **Implication ($p \rightarrow q$)** — "if p then q", false only when p is true and q is false.

bullet **Biconditional ($p \leftrightarrow q$)** — "p if and only if q", true when p and q have the same truth value.

bullet **Truth Tables:** a table listing all possible combinations of truth values of the propositional variables and the resulting truth value of the compound proposition.

bullet **Tautology:** a compound proposition that is always true, regardless of the truth values of its variables (e.g., $p \vee \neg p$).

bullet **Contradiction:** a compound proposition that is always false (e.g., $p \wedge \neg p$).

bullet **Logical Equivalences:** two propositions are logically equivalent if they have identical truth tables (denoted $p \equiv q$). Key laws include Identity, Domination, Idempotent, Double Negation, Commutative, Associative, Distributive, **De Morgan's Laws** ($\neg(p \wedge q) \equiv \neg p \vee \neg q$ and $\neg(p \vee q) \equiv \neg p \wedge \neg q$), and Absorption laws.

1.2 Predicate Logic

Predicate logic extends propositional logic by allowing statements about variables and using quantifiers, enabling more expressive reasoning than fixed true/false propositions.

bullet **Quantifiers:**

bullet **Universal quantifier ($\forall x P(x)$)** — "for all x, P(x) is true".

bullet **Existential quantifier ($\exists x P(x)$)** — "there exists an x such that P(x) is true".

bullet **Nested Quantifiers:** statements using more than one quantifier (e.g., $\forall x \exists y P(x, y)$); the order of quantifiers matters and changes the meaning of the statement.

bullet **Inference rules for predicate logic:** rules such as Universal Instantiation (deduce $P(c)$ from $\forall x P(x)$), Universal Generalization, Existential Instantiation, and Existential Generalization, used to derive valid conclusions from quantified statements.

1.3 Mathematical Proofs (Basic Proof Techniques)

bullet **Direct Proof:** assumes the hypothesis is true and uses logical steps/definitions to directly show the conclusion is true.

bullet **Proof by Contrapositive:** proves "if p then q" by instead proving the logically equivalent statement "if $\neg q$ then $\neg p$ ".

bullet **Proof by Contradiction:** assumes the negation of the statement to be proved, then derives a logical contradiction, concluding the original statement must be true.

bullet **Proof by Mathematical Induction:** used to prove statements about all natural numbers; consists of:

1. **Base Case** — verify the statement holds for the smallest value (usually $n=1$ or $n=0$).

2. **Inductive Step** — assume the statement holds for $n=k$ (inductive hypothesis), and prove it then holds for $n=k+1$.

bullet If both steps succeed, the statement is true for all $n \geq$ base case.

Unit 2.0 — Set Theory and Relation

2.1 Set and Subsets

A **set** is a well-defined collection of distinct objects, called elements.

Subset: set A is a subset of B ($A \subseteq B$) if every element of A is also an element of B.

Sets can be finite or infinite, and may be represented using roster/set-builder notation.

2.2 Operations on Sets

Union ($A \cup B$): all elements that are in A, in B, or in both.

Intersection ($A \cap B$): elements common to both A and B.

Difference ($A - B$): elements in A but not in B.

Complement (A' or A^c): elements in the universal set U that are not in A.

Cartesian Product ($A \times B$): the set of all ordered pairs (a,b) where $a \in A$ and $b \in B$.

2.3 Venn Diagrams and De Morgan's Law

A **Venn diagram** uses overlapping circles within a rectangle (universal set) to visually represent sets and their relationships (union, intersection, difference).

De Morgan's Law for sets: $(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$ — these can be geometrically verified by shading the corresponding regions in a Venn diagram and observing they match.

2.4 Relations and Their Properties

A **relation** R from set A to set B is a subset of the Cartesian product $A \times B$, i.e., a set of ordered pairs (a,b) where $a \in A$ and $b \in B$.

Domain: the set of all first elements (a) appearing in the ordered pairs of R.

Range: the set of all second elements (b) appearing in the ordered pairs of R.

Properties of relations:

Reflexive: $(a,a) \in R$ for every $a \in A$.

Symmetric: if $(a,b) \in R$ then $(b,a) \in R$.

Transitive: if $(a,b) \in R$ and $(b,c) \in R$ then $(a,c) \in R$.

Antisymmetric: if $(a,b) \in R$ and $(b,a) \in R$ then $a = b$.

2.5 Equivalence Relation

A relation that is **reflexive, symmetric, and transitive** simultaneously is called an **equivalence relation**.

An equivalence relation partitions a set into disjoint subsets called **equivalence classes**, where all elements within a class are related to each other.

Applications: equivalence partitioning in software testing, grouping data into categories, modular arithmetic (congruence relations).

Unit 3.0 — Combinatorics

3.1 Basic Counting Principles

- Sum Rule:** if a task can be done in one of two ways, with m choices for the first way and n choices for the second (and they don't overlap), the total number of ways is $m + n$.
- Product Rule (Fundamental Counting Principle):** if a task consists of a sequence of k independent steps, with $n_1, n_2, \dots, n_k$ choices for each step respectively, the total number of ways to complete the task is $n_1 \times n_2 \times \dots \times n_k$.

3.2 Permutations and Combinations

- Permutation:** an arrangement of objects in a specific **order**. The number of permutations of n distinct objects taken r at a time is $P(n,r) = n!/(n-r)!$.
- Combination:** a selection of objects where **order does not matter**. The number of combinations of n distinct objects taken r at a time is $C(n,r) = n!/[r!(n-r)!]$.
- Key distinction: use permutations when arrangement/order matters (e.g., arranging letters in a word), and combinations when only the selection matters (e.g., choosing a committee).

3.3 Pigeonhole Principle

The **Pigeonhole Principle** states that if n items are placed into m containers, and $n > m$, then at least one container must hold more than one item.

- A generalized form states that if n items are placed into m containers, at least one container must hold at least $\lceil n/m \rceil$ items.
- Used to prove existence results (e.g., guaranteeing duplicates, collisions, or shared properties) without needing to construct them explicitly — used to find IP address collisions, duplicate entries in a database, etc.

3.4 Binomial Theorem

The **Binomial Theorem** gives the expansion of $(x + y)^n$ as a sum: $(x+y)^n = \sum_{k=0}^n C(n,k) \cdot x^{n-k} \cdot y^k$.

- The coefficients $C(n,k)$ are called **binomial coefficients**, and can be arranged in **Pascal's Triangle**.
- Applications: computing probabilities in binomial distributions (e.g., probability of outcomes in binary classification problems).

3.5 Generating Functions

A **generating function** represents a sequence of numbers $(a_0, a_1, a_2, \dots)$ as the coefficients of a power series $G(x) = a_0 + a_1x + a_2x^2 + \dots$

- Generating functions transform counting/combinatorial problems into algebraic problems (manipulating polynomials/power series), making it easier to solve recurrence relations and counting problems.
- Applications: solving recurrence relations, analyzing algorithm performance, and counting problems in computer science (e.g., probability generating functions in randomized algorithms).

Unit 4.0 — Graph Theory

4.1 Basic Concepts and Definitions

A **graph** $G = (V, E)$ consists of a set of vertices V and a set of edges E connecting pairs of vertices, used to model pairwise relationships between objects.

4.2 Types of Graph and Degree of Vertex

- bullet **Simple graph**: no loops or multiple edges between the same pair of vertices.
- bullet **Multigraph**: allows multiple edges between the same pair of vertices.
- bullet **Directed/Undirected graph**: edges have/do not have direction.
- bullet **Degree of a vertex**: the number of edges incident on that vertex (for directed graphs, split into in-degree and out-degree). By the **Handshaking theorem**, the sum of degrees of all vertices equals twice the number of edges.

4.3 Subgraph and Isomorphic Graphs

- bullet **Subgraph**: a graph formed from a subset of the vertices and edges of another graph.
- bullet **Isomorphic graphs**: two graphs are isomorphic if there exists a one-to-one correspondence between their vertex sets that preserves adjacency (i.e., they have the same structure, possibly just relabeled/redrawn differently).

4.4 Walks, Paths, Cycle

- bullet **Walk**: a sequence of vertices and edges where consecutive vertices are connected by an edge (vertices/edges may repeat).
- bullet **Path**: a walk in which no vertex is repeated.
- bullet **Cycle**: a closed walk (starts and ends at the same vertex) in which no vertex (other than the start/end) is repeated.

4.5 Eulerian Graph and Its Applications

An **Eulerian graph** contains an **Eulerian circuit** — a closed walk that traverses every edge of the graph exactly once.

- bullet A connected graph has an Eulerian circuit if and only if every vertex has an even degree.
- bullet Applications: route planning problems such as the classic "Bridges of Königsberg" problem, postal delivery routes, and circuit design where every connection must be tested exactly once.

4.6 Connectivity and Components

- bullet **Connectivity**: a graph is **connected** if there is a path between every pair of vertices.
- bullet **Connected component**: a maximal connected subgraph — a disconnected graph consists of two or more such components.
- bullet **Vertex/Edge connectivity**: the minimum number of vertices/edges that must be removed to disconnect the graph, an important measure of network robustness.

Unit 5.0 — Introduction to Fuzzy Set Theory

5.1 Basics of Fuzzy Set Theory

Unlike **classical (crisp) set theory**, where an element either fully belongs (1) or does not belong (0) to a set, **fuzzy set theory** allows **partial membership**, represented by a value between 0 and 1. This makes fuzzy sets suitable for modeling uncertainty, vagueness, and imprecision found in real-world data (e.g., "warm" temperature, "tall" person).

5.2 Membership Functions and Degrees of Membership

A **membership function** $\mu_A(x)$ defines the degree to which element x belongs to fuzzy set A , with $\mu_A(x) \in [0, 1]$.

- Common shapes include **triangular**, **trapezoidal**, and **Gaussian** membership functions.
- Example: a triangular membership function for "warm" temperature might rise from 0 at 15°C to 1 at 25°C and fall back to 0 at 35°C.

5.3 Fuzzy Set Theoretic Operations

Fuzzy sets support operations analogous to classical sets, generalized using membership values:

- Union:** $\mu_{(A \cup B)}(x) = \max(\mu_A(x), \mu_B(x))$.
- Intersection:** $\mu_{(A \cap B)}(x) = \min(\mu_A(x), \mu_B(x))$.
- Complement:** $\mu_{A'}(x) = 1 - \mu_A(x)$.
- Extension Principle:** a general method to extend classical mathematical operations/functions to operate on fuzzy sets by considering the membership degrees of the inputs.

5.4 Fuzzy Propositions and Truth Values

A **fuzzy proposition** is a statement whose truth value is not strictly true or false, but a degree of truth in the interval $[0,1]$, reflecting the imprecision in natural language statements (e.g., "the room is somewhat hot" might have a truth value of 0.7).

- Fuzzy logic uses these graded truth values, along with fuzzy operators (AND = min, OR = max, NOT = $1-x$), to perform approximate reasoning — widely applied in control systems, AI, and decision-making under uncertainty.